

Project Design Phase-II

Technology Stack (Architecture & Stack)

Project: Freelancer Web Application (MERN Stack)

Date	10 April 2026
Team ID	Your Team ID
Project Name	Freelancer Web Application
Maximum Marks	4 Marks

Technical Architecture

The Freelancer Web Application follows a 3-tier architecture consisting of frontend (React), backend (Node.js & Express), and database (MongoDB). This architecture ensures scalability, performance, and maintainability.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	User interaction layer	HTML, CSS, React.js
2	Backend Logic	Business logic & APIs	Node.js, Express.js
3	Authentication	Login & security	JWT, bcrypt
4	Database	Data storage	MongoDB
5	Cloud DB	Hosted DB	MongoDB Atlas
6	File Storage	Media storage	Cloudinary / AWS S3
7	Messaging	Real-time chat	Socket.io
8	Payment	Transactions	Razorpay / Stripe
9	External APIs	Third-party services	Nodemailer, Payment APIs
10	Deployment	Hosting	Vercel, Render
11	Version Control	Code management	Git, GitHub

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open Source	Frameworks used	React, Node, Express
2	Security	Data protection	JWT, bcrypt, HTTPS
3	Scalability	Handles growth	MERN architecture
4	Availability	System uptime	Cloud hosting
5	Performance	Fast response	Optimized APIs
6	Maintainability	Easy updates	MVC pattern
7	Compatibility	Cross-platform	Responsive design

Conclusion

The selected technology stack ensures that the Freelancer MERN application is scalable, secure, and efficient.